# Huawei Access Node BAS-ACC-73 Optical Transport Failure Resolution SOP

**Node:** BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7512

## Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node BAS-ACC-73 at Basavanagudi POP is a critical failure domain that can occur due to the intrinsic nature of optical transmission protocols. The SONET/SDH ring topology employed in this node is prone to failure when the SFP/XFP transceivers degrade, leading to signal attenuation, or when fiber cuts occur, causing signal loss. Additionally, Rx/Tx power loss can arise from faulty or malfunctioning optical amplifiers or regenerators. Furthermore, the Huawei Access Node's reliance on optical transport protocols, such as OC-48 or OC-192, makes it susceptible to these types of failures. In the event of an ALARM_7512, it is essential to promptly identify and rectify the root cause to prevent network-wide disruptions and ensure continuous service availability.

## Pre-requisite Checks

- Confirm the ALARM_7512 is indeed related to the Optical Transport Failure Domain

- Verify the redundant link status to ensure it is not the cause of the failure

- Dispatch remote hands to the Basavanagudi POP to provide physical access to the node

- Ensure all necessary safety precautions are taken before commencing diagnostic and resolution procedures

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP Transceiver Degradation | High |
| Fiber Cut | Medium |
| Rx/Tx Power Loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display evn evn-index 1, Expected Output: Evn Index: 1, Evn Type: Optical Transport, Description: This command displays the EVN (Enhanced Virtual Network) index and type, confirming the node's role in the optical transport network. It helps identify if the failure is related to the EVN or not.

Command: display optical transceiver, Expected Output: Transceiver Type: SFP/XFP, Transceiver Status: Degraded, Description: This command displays the SFP/XFP transceiver type and status, helping to determine if the transceiver is the root cause of the failure.
```

```
Command: display fiber status, Expected Output: Fiber Status: Cut, Fiber Location: Basavanagudi POP, Description: This command displays the fiber status and location, confirming if the fiber cut is the cause of the failure.
```

## Resolution Steps

Step 1: Step: Replace the SFP/XFP Transceiver, Command: replace SFP/XFP transceiver , Description: Replace the degraded SFP/XFP transceiver with a new one to restore the optical transport link. Ensure the new transceiver is properly seated and secured.

Step 2: Step: Repair or Replace the Fiber, Command: repair/replace fiber , Description: Repair or replace the cut fiber to restore the optical transport link. Ensure all necessary safety precautions are taken before commencing the repair or replacement procedure.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: adjust rx-tx-power , Description: Adjust the Rx/Tx power levels to ensure proper signal integrity. This may involve adjusting the optical amplifier or regenerator settings.

## Post-Check Verification

```
Command: display optical transceiver, Expected Output: Transceiver Type: SFP/XFP, Transceiver Status: Normal, Description: Verify the SFP/XFP transceiver status to ensure it is functioning normally.

Command: display fiber status, Expected Output: Fiber Status: Normal, Fiber Location: Basavanagudi POP, Description: Verify the fiber status to ensure it is functioning normally.

Command: display evn evn-index 1, Expected Output: Evn Index: 1, Evn Type: Optical Transport, Description: Verify the EVN index and type to ensure the node is functioning correctly in the optical transport network.
```

## Roll-back Procedures

- Step: Rollback to Previous Configuration, Command: undo replace SFP/XFP transceiver , Description: Rollback to the previous configuration by undoing the SFP/XFP transceiver replacement. Ensure all necessary safety precautions are taken before commencing the rollback procedure.

- Step: Restore Original Fiber Configuration, Command: undo repair/replace fiber , Description: Restore the original fiber configuration by undoing the repair or replacement procedure. Ensure all necessary safety precautions are taken before commencing the rollback procedure.

- Step: Restore Original Rx/Tx Power Levels, Command: undo adjust rx-tx-power , Description: Restore the original Rx/Tx power levels by undoing the adjustment. Ensure all necessary safety precautions are taken before commencing the rollback procedure.